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Design and structure of protein cages based on helical fusion and machine learning

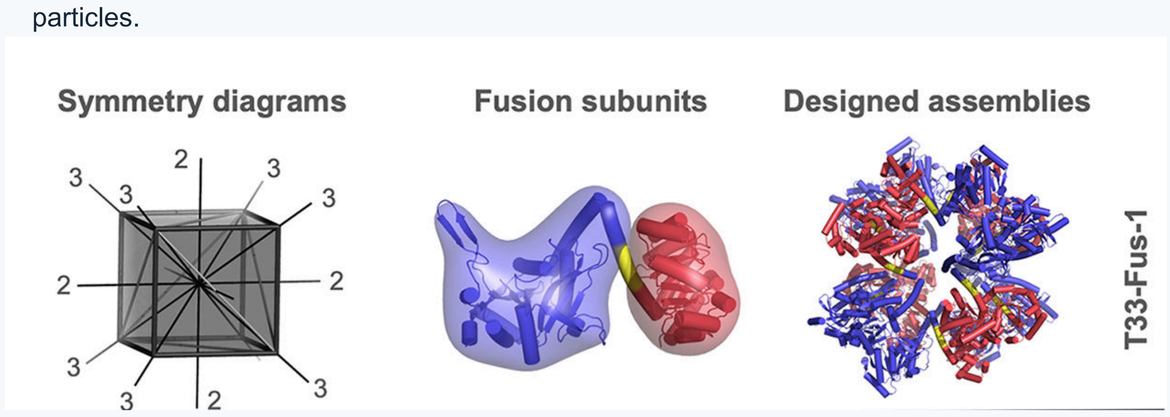
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Fusing two homotrimers via a single continuous α-helix generates a self-assembling tetrahedral cage without requiring de novo interface design. However, closure demands the 3-fold axes meet at exactly 70.5°, an angle a rigid linker cannot achieve. By analysing high-resolution structures, we mapped preferred modes of helical deformation, finding that helices bend readily but strongly resist stretching. We scored candidate fusions based on their structural deviation from these natural modes. From 15 filtered designs, three successfully assembled into cages, and cryo-EM reconstructions at 3.0–3.9 Å resolution precisely reproduce the predicted helical bend.

1 THE FUSION PROBLEM

Rigidly fusing two homotrimers via an α-helix locks their 3-fold axes, driving 12 subunits to self-assemble into a tetrahedral T33 cage. Historically, the continuous helical linker has been a vulnerability: unmodelled deformation often results in distorted or incorrectly symmetric particles.



Two trimer types (red, blue) on alternating vertices of a cube, bridged by one helix (yellow).
Fusion needs no new interface: the cage stays single-component.

2 A GEOMETRY-AWARE PIPELINE

1 Mine trimers — PDB homotrimers with a straight end helix.

2 Fuse pairs — Merge both terminal helices into one.

3 Check geometry — Keep pairs whose C3 axes meet at 70.5°.

4 Score the bend ★ — Rank how naturally the bend conforms to 70.5°. *key advance*

5 Filter — Drop clashes and open cavities → 15 designs.

6 Sequence design — ProteinMPNN redesigns only the junction.

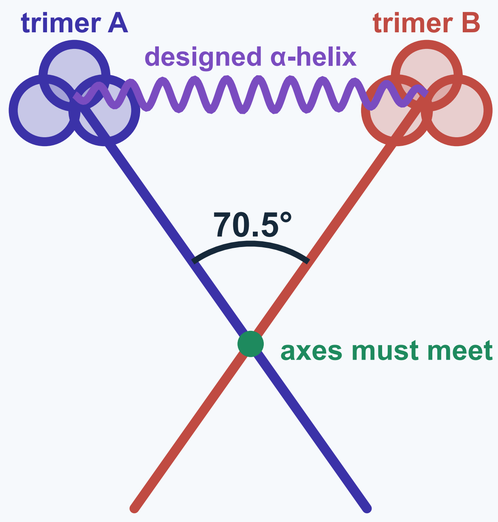
7 Express & screen — E. coli, IMAC and SEC → 3 winners.

8 Cryo-EM — 3.0–3.9 Å structures with T symmetry.

Atomically accurate trimer-trimer T33 cages:

ProteinMPNN changed only 10–12 positions of the starting fusion model.

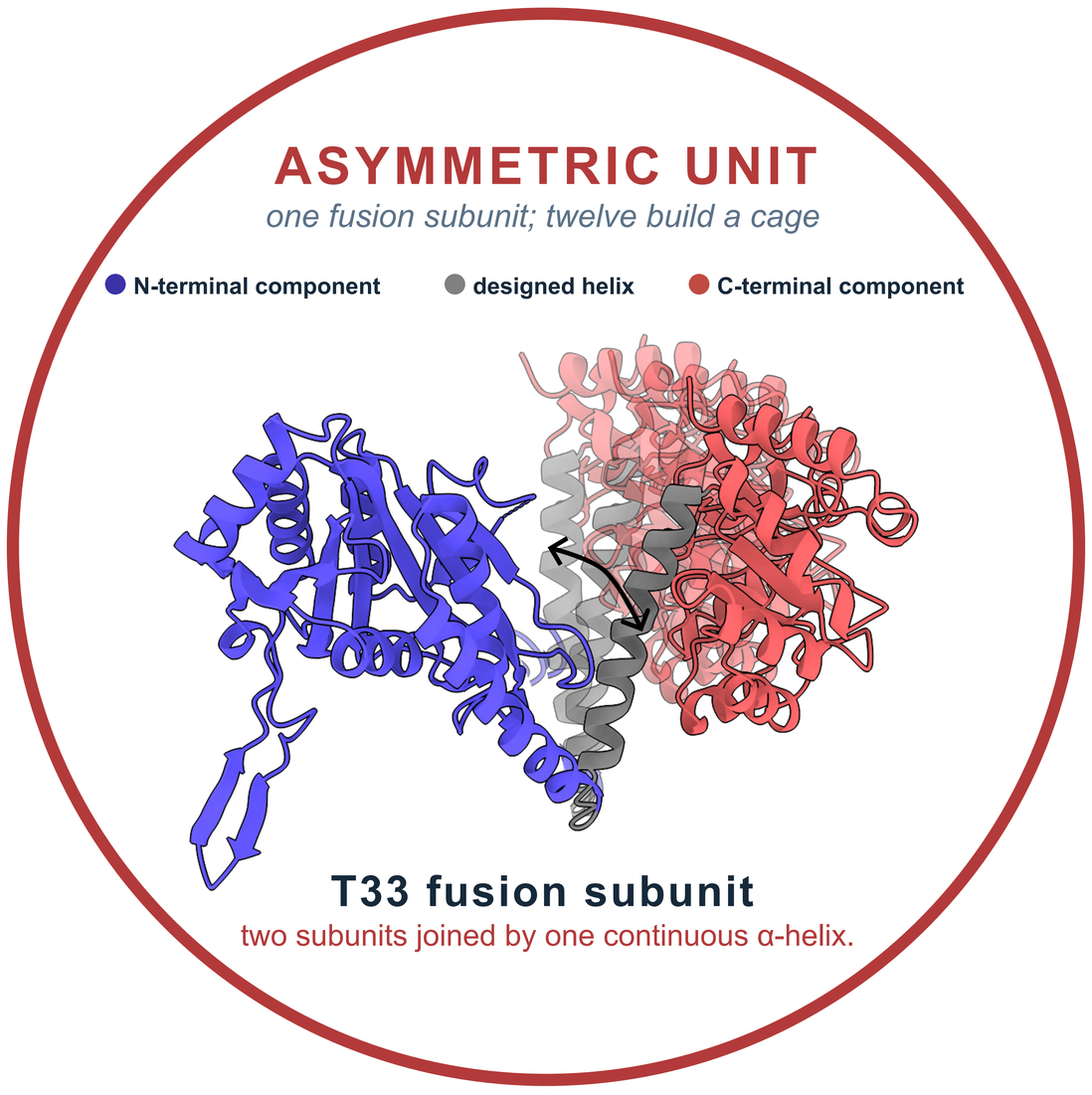
3 CLOSURE CONDITION



× 12 subunits → closed tetrahedral cage

Tetrahedral symmetry requires the two 3-fold axes to intersect at exactly arccos(1/3) = 70.5°. Angle alone is insufficient; if the axes fail to intersect, there is no cage centre and the polyhedron cannot close.

Because real building blocks inevitably deviate from perfect geometry, the linking helix must absorb this structural strain. Success depends not on perfect rigid-body alignment, but on whether the required helical deformation aligns with naturally favoured bending modes.



4 SCORING THE BEND

E = energetic cost of required deformation
d_i = required geometric deviation · λ_i = observed variance

$$E = \Sigma\, d_i^2 / \lambda_i$$

bend λ = 0.44 (observed)

bend (90°) λ = 0.41 (observed)

twist λ = 0.03 (observed)

stretch λ ≈ 0 (never seen)

By aligning seven-residue helices from high-resolution (<1.5 Å) structures on their first three residues, we defined standard variation modes (λ) for the far end.

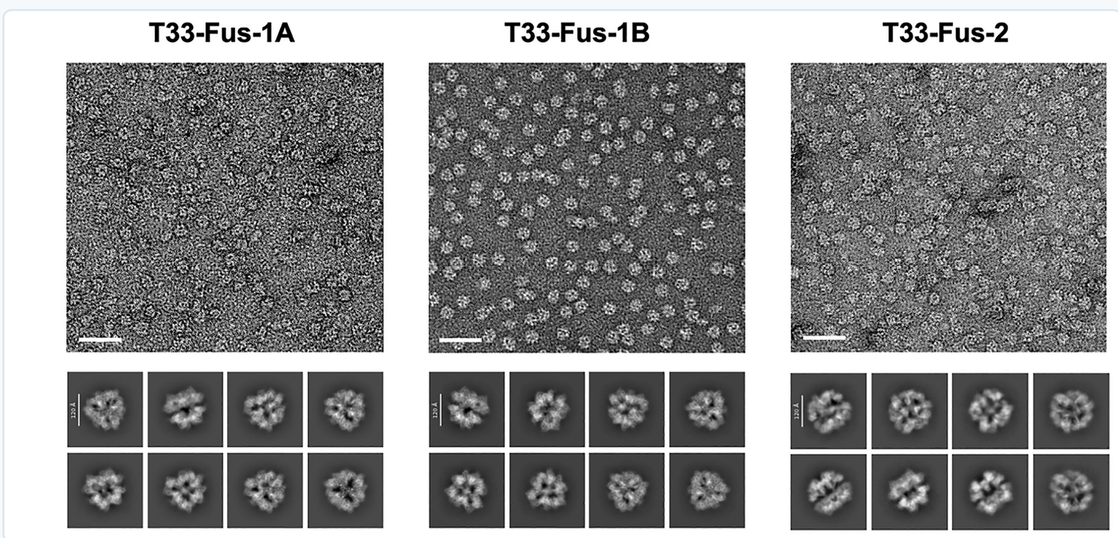
Three primary modes account for 90 % of observed natural helical flexibility. Notably, axial stretching is virtually non-existent in nature, effectively eliminating any design that requires it.

For the same 0.3 Å correction across two directions:

bending **0.3² / 0.44 = 0.20** (energetically cheap — accept)

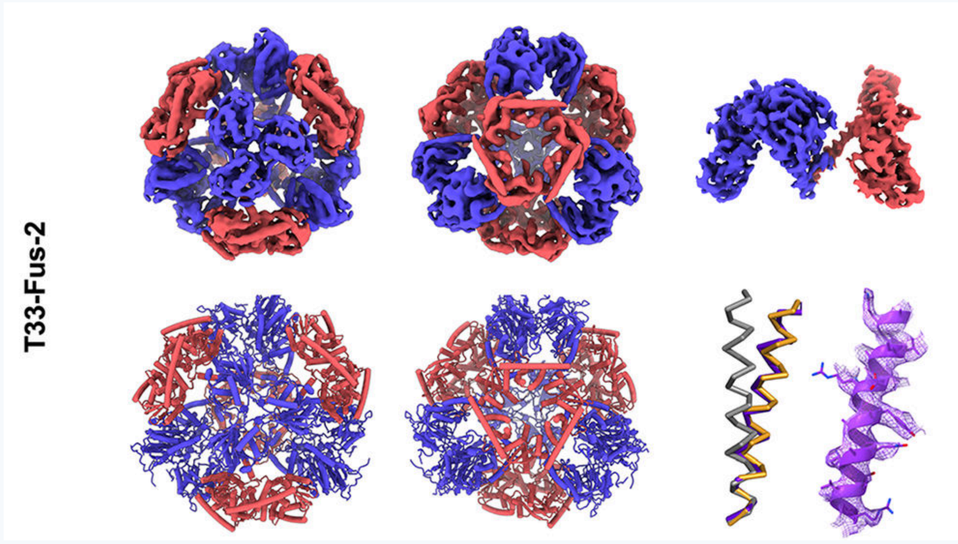
stretch **0.3² / 0.0008 = 112** (≈550× costlier — reject)

5 CAGES THAT ASSEMBLE



15 designs → 45 constructs expressed in E. coli; five eluted as the 12-subunit particle and three went forward. Negative-stain TEM shows monodisperse ≈150 Å cages with no aggregates or fibrils; native masses are 633 kDa (1A/1B) and 638 kDa (2).

6 CRYO-EM CONFIRMS THE DESIGN



Maps at 3.0, 3.9, and 3.8 Å confirm the architecture: RMSD to the design model is 2.0 and 4.9 Å, representing 1.2% and 1.9% of the 170 Å particle. T33-Fus-2 deviates most from an ideal helix (grey), yet its engineered bend (orange) matches the experimental structure (purple).

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EMD-59307, 59308, 59315–59318 · PDB 33AR, 33AS, 33AT, 33AU, 33AV, 33AW
Raw data: EMPIAR-13888, 13889, 13890

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